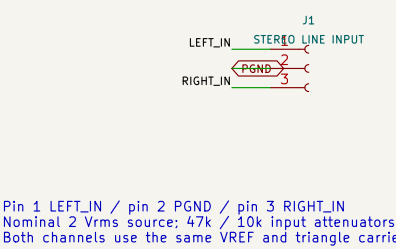


STEREO LINE INPUT



IC ALLOCATION / REV 1

U1A: LEFT input follower
U1B: RIGHT input follower
U3A: VREF_2VS buffer
U3B: shared triangle integrator

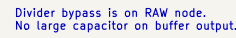
U7A: oscillator Schmitt comparator
U7B: LEFT PWM comparator
U8A: RIGHT PWM comparator
U8B: spare: inputs tied, output open

U4 / U5: LEFT inverter / dual Schmitt buffer
U9 / U10: RIGHT inverter / dual Schmitt buffer
U11: +24V to +12V/DC TP57AA901
U12: +24V to +5V/DC TP57AA901

U2 / U6: LEFT/RIGHT UCC27301A gate drivers
U102 / U106: RIGHT UCC27301A gate drivers

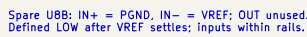
Total: 14 IC packages, 8 power MOSFETs.

3. SHARED 400 kHz OSCILLATOR



Positive-feedback Schmitt: triangle $\rightarrow$ 10k $\rightarrow$ IN+; OUT $\rightarrow$ 49k9 $\rightarrow$ IN+; IN- = VREF
Inverting integrator: square $\rightarrow$ 3k09 $\rightarrow$ IN-; INf from OUT to IN-; IN+ = VREF
Ideal 403.722 kHz; 1.9993 to 3.0014 V at nominal 5.0007 V rail, Tune / measure.

DESIGN / COMMISSIONING NOTES



Dyn-Loop Rep I/O: no audio feedback or protection.
Target 30 W/channel into 8 ohms at +24 V.
Power target and timing require bench verification.

Input attenuation = $-10\text{V} / (\sqrt{2}) \times 100 = 0.707\text{V}_{RMS}$.
This comes from the fact that $\sqrt{2} \times 0.707\text{V}_{RMS}$
= RMS input. 195 kHz from $(47k) / (100k) \times 100p$.
 2 V_{RMS} level gives $0.351\text{ V}_{RMS} / 0.446\text{ V}_{pk}$ modulation.

Non-Inverting Summing (+ Inverting) Integrator:
 $V_{in1} = \text{Inr } (10k)$
 $495k / (10k \times 300) = 1.65 \times 10^3 = 1.65 \times 222\text{ kHz}$
Computer aided, hysterisis and op-amp dynamics
change after each iteration of *priced*, *measure* and *tune*.

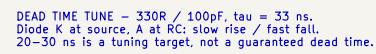
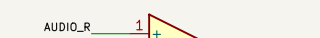
Dead time is proportional: RC tau = 33 ns.
The 33 ns is due to the factors of *priced*, *measure* and *tune*.
Threshold sense, propagation slew, device source.
Belt, clitics, diodes, capacitors, gates, etc. are all
there. Measure all eight gates before selecting final timing.

Both channels default DISABLED (300 pulldowns).
ENAB 1 pulls up terminal after standalone enabling.

PFGM is the single global board-wide -0 V reference.
Speaker outputs float to ground either terminal.
But capacitors are shared; local source bypass splits.

Metric SMD PCB/Land patterns assigned, so placement,
placement, placement, placement, placement means
before PCB layout / manufacture.

5. RIGHT INPUT + PWM + DEAD TIME



DEAD TIME TUNE = $330R / 100pF$, $\tau = 33$ ns.
Diode K at source, A at RC; slow rise / fast fall.
20-30 ns is a tuning target, not a guaranteed dead time.



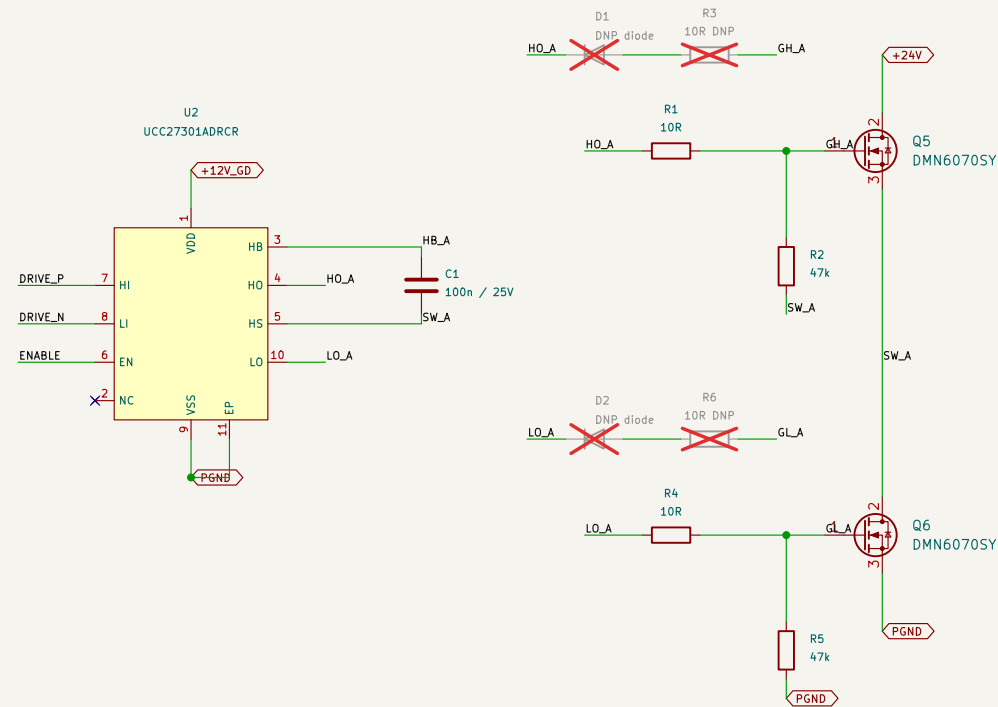
File: PowerOutput_R_kload.sch

Enable defaults LOW. Fit DNP 1k only for standalone enable.
100n here belongs to U4; Schmitt package bypass is in IC POWER row.

Enable defaults LOW. Fit DNP 1k only for standalone enable.
100n here belongs to U9: Schmitt package bypass is in IC POWER row.

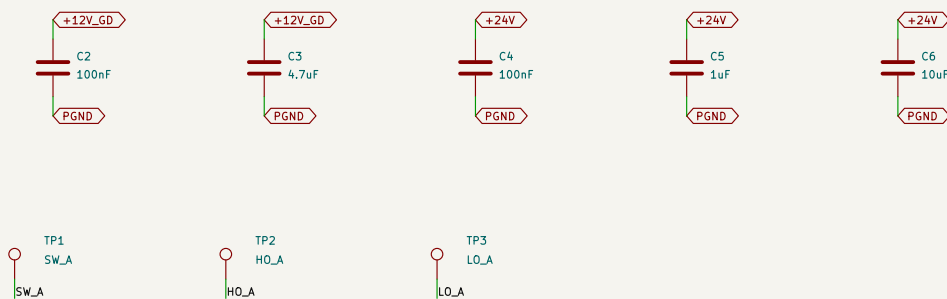
OPA1656 U1 / U3: +12V_GD supply; VREF remains 2.5 V.
DMN6070SY MOSFETs retained in both channels; G=1, D=2/tab, S=3.
DEAD TIME TUNE: gate timing and regulator thermals require bench validation

HALF BRIDGE A



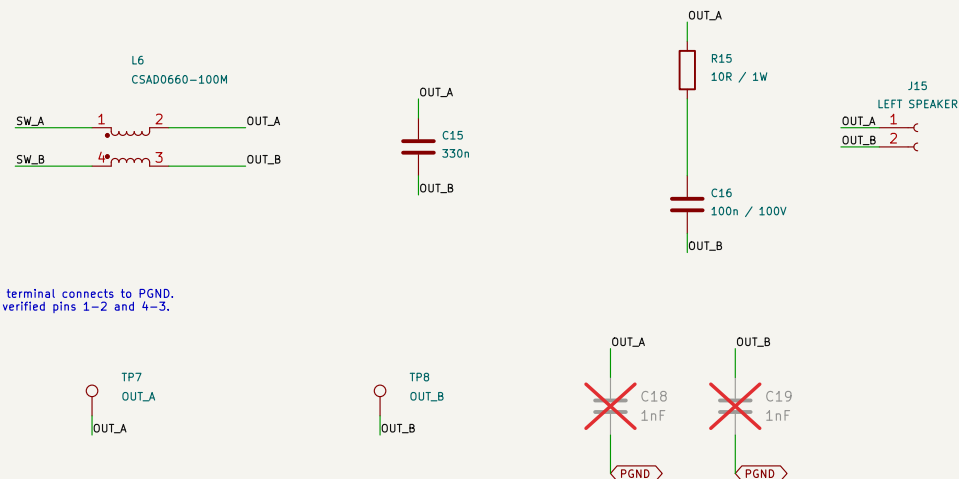
Driver bypass: at VDD/VSS

Local 24 V power loop ceramics



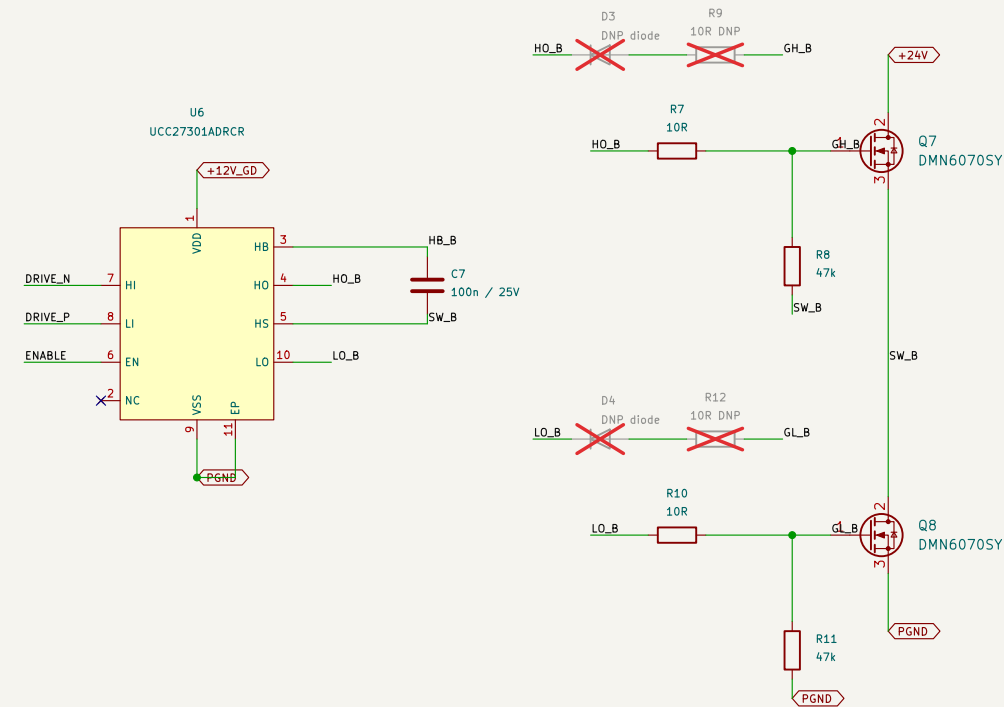
HS / VSS: Kelvin returns to respective MOSFET sources.
Complementary DRIVE_P / DRIVE_N include external dead time.
DNP diode cathode faces driver: optional faster turn-off path.

DIFFERENTIAL OUTPUT / 8 OHM SPEAKER



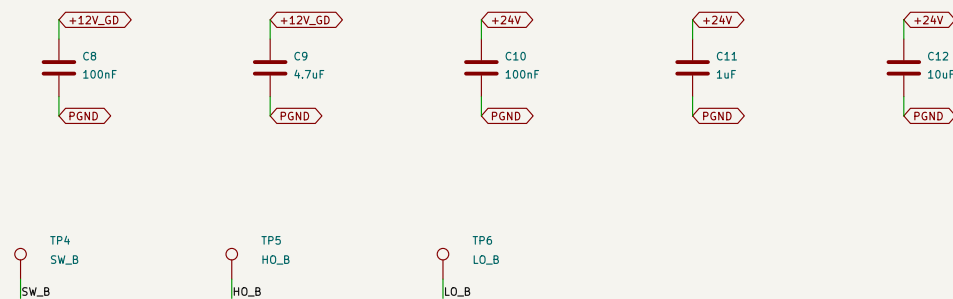
Speaker floats: neither terminal connects to PGND.
L6: 10uH per winding; verified pins 1-2 and 4-3.

HALF BRIDGE B



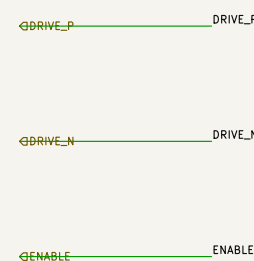
Driver bypass: at VDD/VSS

Local 24 V power loop ceramics

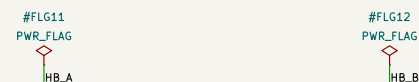


HS / VSS: Kelvin returns to respective MOSFET sources.
Complementary DRIVE_P / DRIVE_N include external dead time
DNP diode cathode faces driver: optional faster turn-off path.

CHANNEL CONTROL / SHARED SUPPLIES



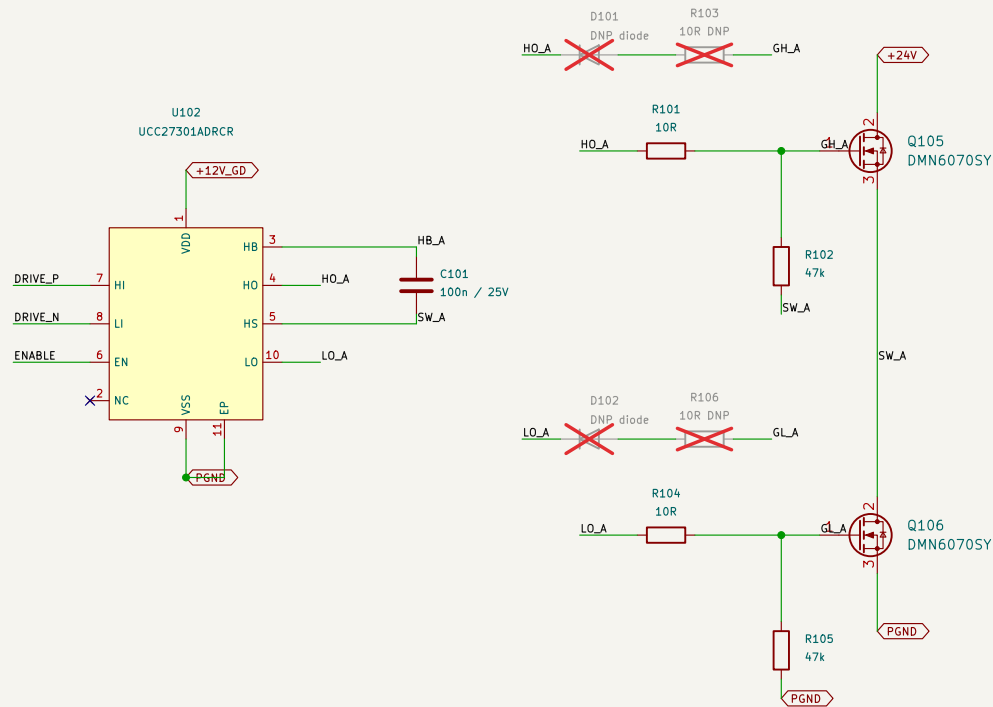
Bridge A: HI = DRIVE_P, LI = DRIVE_N.
 Bridge B: HI = DRIVE_N, LI = DRIVE_P.
 ENABLE drives both EN pins. Defaults LOW on main.
 +24V / +12V_GD / PGND are global.
 Speaker and switch nets remain local to this channel.
 PGND is the single board-wide 0 V reference.



HB flags represent charging through each UCC27301A internal bootstrap diode.

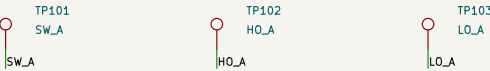
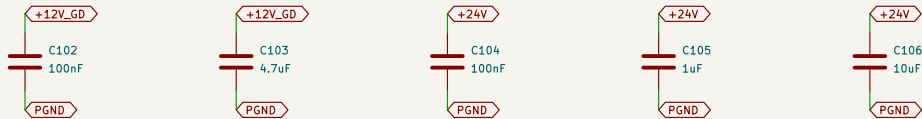
Sheet: /LEFT OUTPUT/ File: PowerOutput_L.kicad_sch		
Title: LEFT BTL power output		
Size: A3	Date:	Rev: Stereo Rev 1
KiCad E.D.A. 10.0.5		Id: 2/3

HALF BRIDGE A



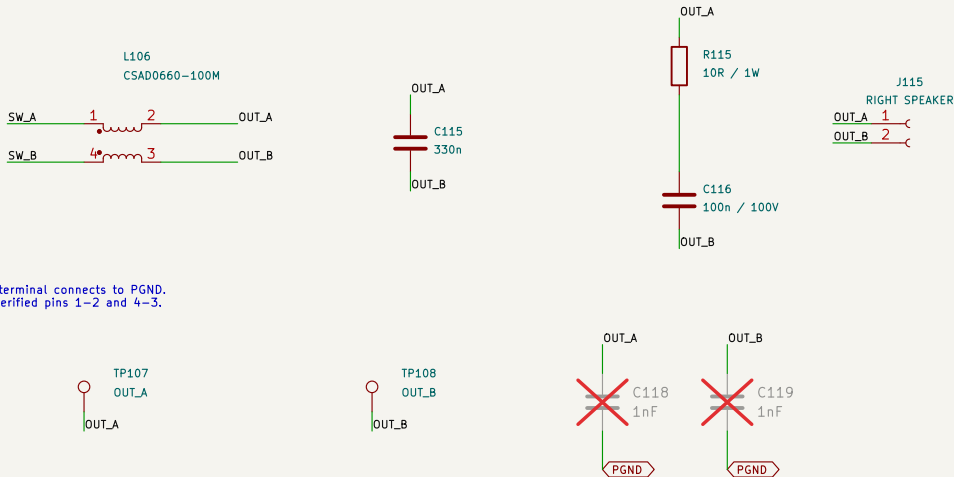
Driver bypass: at VDD/VSS

Local 24 V power loop ceramics

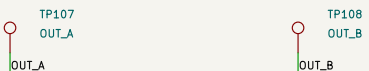


HS / VSS: Kelvin returns to respective MOSFET sources.
Complementary DRIVE_P / DRIVE_N include external dead time.
DNP diode cathode faces driver: optional faster turn-off path.

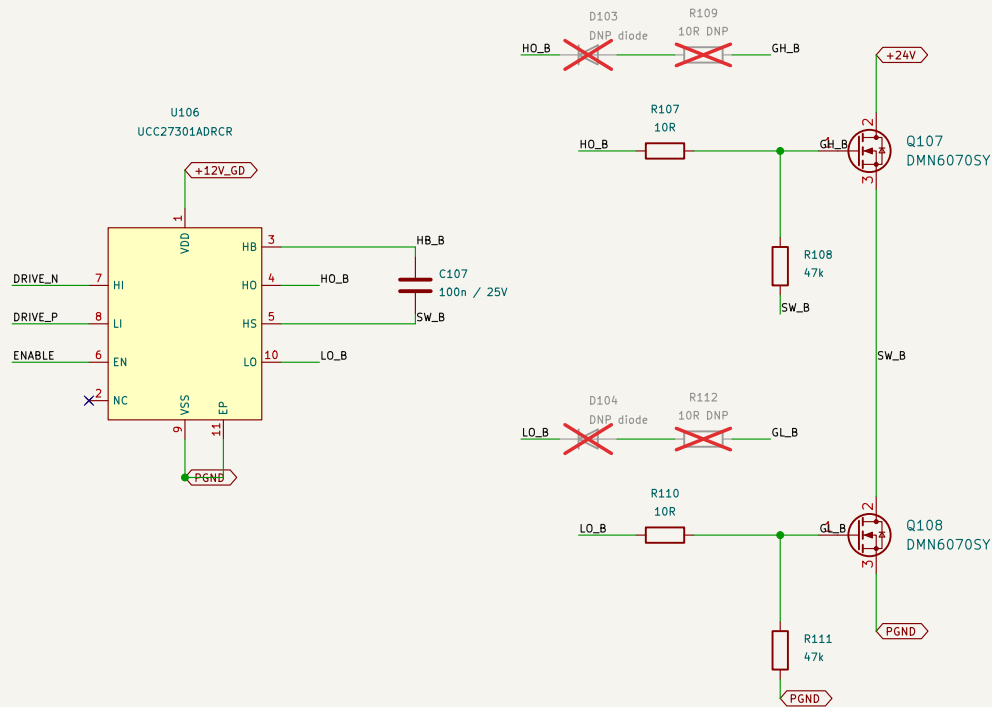
DIFFERENTIAL OUTPUT / 8 OHM SPEAKER



Speaker floats: neither terminal connects to PGND.
L6: 10uH per winding; verified pins 1-2 and 4-3.

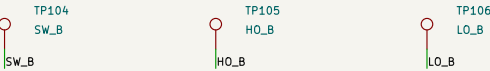
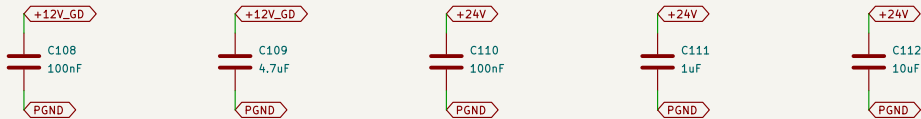


HALF BRIDGE B



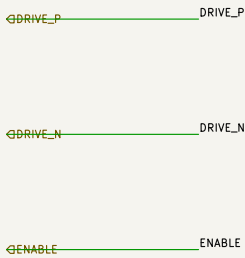
Driver bypass: at VDD/VSS

Local 24 V power loop ceramics



HS / VSS: Kelvin returns to respective MOSFET sources.
Complementary DRIVE_P / DRIVE_N include external dead time.
DNP diode cathode faces driver: optional faster turn-off path.

CHANNEL CONTROL / SHARED SUPPLIES



Bridge A: HI = DRIVE_P, LI = DRIVE_N.
Bridge B: HI = DRIVE_N, LI = DRIVE_P.
ENABLE drives both EN pins. Defaults LOW on main.
+24V / +12V_GD / PGND are global.
Speaker and switch nets remain local to this channel.
PGND is the single board-wide 0 V reference.



HB flags represent charging through each UCC27301A internal bootstrap diode.

Sheet: /RIGHT OUTPUT/ File: PowerOutput_R.kicad_sch		
Title: RIGHT BTL power output		
Size: A3	Date:	Rev: Stereo Rev 1
KiCad E.D.A. 10.0.5		Id: 3/3